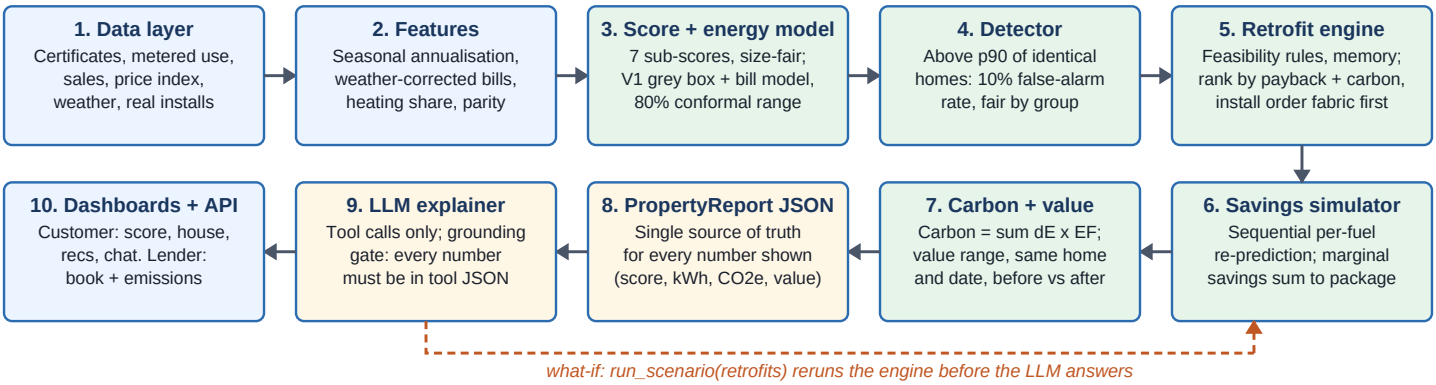
Green Home Score & Retrofit Recommendation Engine

A deterministic engine that scores a home, recommends feasible retrofits in the right order and estimates savings, carbon and value, validated on real metered homes and real installs, with an assistant that can only explain the engine's numbers

1. Solution

For each home the engine builds one **PropertyReport**, the single source of truth that the customer dashboard, the lender portfolio view, the API and the assistant all read and never recompute: a 0-100 Green Home Score with its seven weighted parts, expected electricity and gas use with an 80% range, a 'worse than similar homes' check, feasible retrofits in priority order (fabric, controls, heating, generation) with marginal per-fuel savings, bill saving and payback, carbon from official 2026 factors, and an indicative property value range before vs after. It runs on **542,711 real homes** from energy performance certificates in three contrasting local authorities (urban terraces, rural off-gas cottages, inner-city flats), joined to 175,457 real sales (92.2% match, only certificates lodged before each sale). Models learn from a 50,000-home national sample of **metered** annual gas and electricity, and are checked against **4,791 real loft, cavity-wall and solar installs** with metered use two years before and after. Synthetic data fills only what no public source has (monthly bills, customer IDs, smart controls, loan-to-value) and is never used to report accuracy.



2. Approach

Data, models and training

Model	Predicts	Method	Training data and split
Energy V1 (shipped)	Annual gas and electricity from the home's description	Heat-loss physics x LightGBM residual (Huber loss, Optuna-tuned); physics elasticity fitted on real installs	50,000 metered homes (2023); fixed 70/15/15 split stratified by region x fuel; 5-fold CV on train
Energy V2	Same	LightGBM direct, same tuning, monotone constraints	Same split and folds
Bill model	Next year's use for a customer with bills	LightGBM on log(this year / last year), L1 loss; one or two years of history	Same homes, 2021-2023
80% ranges	p10 / p90 for every prediction	Quantile LightGBM + split-conformal correction (CQR)	Fitted on train, calibrated on validation
Value	Price range today, before vs after	LightGBM on log price indexed to the valuation date, monotone in rating; CQR range	151,587 real resales 2018-2025; 106k / 23k / 22k by customer; CV grouped by home
Rating premium	% value per rating band	Hedonic log-linear, 259 postcode-sector fixed effects, clustered errors	Train-split sales
Rating after retrofit	SAP score after a measure	Linear regression on fabric fields	376,769 real certificates (train customers)

- **Inputs and leakage rules.** One feature function serves training and deployment, using only what a certificate also gives: property type, age band, floor area, rating band, region, heating fuel, deprivation quintile and retrofit flags (tested to match). Targets are log annual kWh per fuel. Installs count only before the target year, sales use only the certificate lodged before the sale, and splits are by home or customer, fixed and versioned. Every choice (objective, Optuna search, monotone constraints, features, LightGBM vs CatBoost) was made on 5-fold CV of the train split; validation calibrates the ranges; the test split was scored once.
- **Deterministic engine (non-ML).** Monthly bills are annualised with seasonal weights and flagged when coverage is low; a per-home degree-day regression (PRISM) weather-corrects them and splits heating from base load. Heat loss uses RdSAP-style U-values and geometry with SAP hot-water and appliance formulas. The score combines seven

monotone sub-scores with the fixed weights; feasibility rules come from a sourced retrofit catalogue; the savings simulator applies measures in order (fabric, controls, heating, solar), each re-predicting from what the last left, per fuel; carbon uses official 2026 factors and bills the current price cap. Every constant lives in config with its source. The assistant is Gemini 2.5 Flash with tool calling behind a grounding gate; FastAPI and Streamlit serve the API and dashboards.

What is new

- **A grey-box energy model calibrated on real retrofits, not on its own equations.** V1 computes heat loss per element (roof, walls, windows, floor, air) times regional degree days, then log(use) = $\beta \cdot$ log(physics) + GBDT(home description). Steady-state physics over-predicts metered gas about 2× (the prebound effect: real homes are heated less than assumed), and over-predicts retrofit savings 3×. We fit the elasticity $\beta = 0.38$ on the observed installs in the train split only, so a retrofit's effect is (physics after / before)$^{\beta}$: smooth, never the wrong sign, with diminishing returns and interactions from the physics, and a per-element heat-loss share for the house diagram.
- **Non-circular evaluation.** Headline accuracy comes only from real metered homes and real installs. Synthetic bills carry hidden occupancy and behaviour the engine never sees, and are used only to test the pipeline and the parts real data can't (controls, heat pumps, glazing), never to report accuracy.
- **Two prediction modes, and a model that learns from the customer's own bills.** *Similar homes* (description only) for prospects and for the detector, which must not see the home's own use; *this home* for existing customers, a GBDT on the year-on-year change log(this year / last year). Since the bill alone carries most of the signal, it is the reference to beat. Each year of history is weather-corrected to a typical year with the home's own degree-day fit, as the national metered data is; the same fit splits each home's heating from hot water and cooking for the savings simulator.
- **Honest uncertainty with stated error rates.** Conformalised quantile models (CQR) give every prediction an 80% range. The 'worse than similar homes' flag fires above the 90th percentile for identical homes, so its false-alarm rate is 10% by construction, and it is checked in every region, type, age, rating and deprivation group.
- **A size-fair score.** Raw kWh/m^2 scored small homes 23.5 points below large ones on real metered use, penalising flats and smaller (often lower-income) homes for their size. The consumption part now scores actual / expected for similar homes: the gap falls to 3.6 points. Every mapping is monotone and in config with a source.
- **A value uplift that compares homes on the same streets.** Without location controls the rating premium comes out **backwards** (badly rated homes are often expensive central period houses). A hedonic model with 259 postcode-sector fixed effects and clustered errors gives the expected order (vs D: A/B +1.0%, C +0.7%, E -2.7%, F/G -5.1%), confirmed by a refit on test sales. The report shows the same home on the same date before vs after, as a range that can include 0, labelled an association, never causal.
- **Recommendations ranked by value and checked against hidden truth.** Suitability comes from payback at current capped prices, with a carbon floor and an 'insulate first' rule for heat pumps. The ranking is scored against the synthetic world's hidden true savings, not the engine's own numbers.
- **An assistant that cannot invent numbers.** The LLM only reads tool output (report, scenario rerun, customer memory). A what-if question reruns the engine before the model answers, and a grounding gate checks every number in the answer against the tool JSON, retries once, then falls back to a deterministic template.
- **Customer memory that changes the engine's inputs, not just the chat.** Each customer has a persistent profile the assistant reads and writes through tools. Completed retrofits update the home before the engine runs, so they are never recommended again and savings start from the new state; stated constraints become feasibility rules (a tenant gets no structural works, measures above a stated budget drop to Low); a declined measure drops to Low with the reason shown ('You declined this on 12 Oct') and every rank is recomputed. Numbers still come only from the engine.

3. Features

- **Customer view:** score with points per part; the heat-loss house with each part's fix; what-if toggles that rerun the engine and move the score, savings and house together; ranked recommendation cards (bill saving, payback, carbon, cost, why); grants and green-finance matching with monthly cash flow; the grounded assistant with an audit of every number; 'Mark completed' / 'Not interested' buttons and a three-session memory timeline.
- **3D thermal twin of the home:** an interactive WebGL (Three.js) model built from the home's own description, with a pitched or flat roof and solar panels or a heat-pump unit when present. Roof, walls, windows and floor glow by their share of heat loss from V1's per-element physics; hovering a part shows its share, U-value and area, and it can be rotated and zoomed. Ticking a what-if rebuilds it from the retrofitted home, so the colours change as insulation goes in; a 2D cross-section is the lightweight view.
- **Lender view:** a mortgage book of real homes: score and rating mix, share Poor / Very poor, financed emissions (PCAF, loan-to-value attribution), reduction potential, a 2030 EPC C target simulator and a ranked list of who to contact about green finance.
- **Electricity and gas are tracked separately end to end;** a heat pump's extra electricity is a negative saving, and solar cuts what is bought from the grid, not demand. Missing data is 'unavailable' and flagged, never 0. FastAPI exposes reports, scenarios, portfolio, chat and memory.

4. Results

Energy model

Real metered homes, test split (scored once)	Gas MAE (kWh)	Gas median error	Gas R^2 (log)	Elec MAE (kWh)	Elec median error
Median of similar homes (no-model baseline)	4,177	30.7%	0.25	1,569	37.9%
V1 grey box, description only	**4,018**	**29.6%**	**0.27**	**1,535**	**37.2%**
Last year's bill alone (no-model baseline)	2,005	11.4%	0.67	917	16.0%
V1 with the customer's own bills	**1,953**	**10.9%**	**0.69**	**886**	**15.6%**

- **Accurate where it matters for a lender:** predicted totals for 1,000 homes are within **1.7%** of metered use, gas R^2 is **0.79** across groups of similar homes, and with the customer's own bills the median error is **10.9%**, better than the bill alone on the same held-out homes.
- **Calibrated 80% ranges:** test coverage 79.2% gas and 80.7% electricity, and 76-84% in every region, type, age, rating and deprivation group; the detector flags 10.2% / 9.7% of homes against 10% by design.
- **Bill history:** a second year of bills cuts error a further 1.5% (gas) / 2.7% (electricity) on validation, and weather correction removes a 7% under-forecast after a mild year.

Retrofit savings model

Real installs, test split	Observed	Physics alone	V1	V2 (GBDT)
Cavity wall insulation (225 homes)	-11.4%	-31.5%	**-13.4%**	-3.8%
Loft insulation (319 homes)	-1.3%	-5.3%	**-2.1%**	no loft input

- **Savings match real installs:** one elasticity fitted on train installs brings V1 within about 2 points of the observed change on held-out installs, where physics alone overstates 3-4×. Solar, recalibrated on real installs, predicts -488 vs -483 kWh/yr observed.
- **All seven measures:** on 1,175 synthetic homes that retrofit mid-series, seeing only pre-install bills, the predicted total saving is within **3%** of the hidden truth and the sign is right for 96% of homes and every heat pump and solar install.

Property value model

Price model, real resales, test split	Median error	Within 20%	R^2 (log)	MAE (GBP)
Area median x fixed rating premium (brief's baseline)	32.6%	32.0%	0.17	150,200
Value model (LightGBM, conformal range)	**14.9%**	**62.8%**	**0.76**	**73,900**
Homes sold before: last sale indexed + model	**8.2%**	**82.9%**	**0.87**	**44,100**

- **Price level:** LightGBM trained on 106k real resales matched to the certificate lodged before each sale, monotone in rating, with conformal 80% ranges (test coverage **79.3%**, 78-81% in every area, type and rating band).
- **Rating uplift:** comparing homes on the same streets (259 postcode sectors) gives the premium vs band D: A/B +1.0%, C +0.7%, E -2.7%, F/G -5.1%, confirmed by a refit on test sales. A retrofit's new rating comes from a SAP-score model on 376,769 real certificates (R^2 0.70).

Recommendations and assistant

Ranking of feasible retrofits (1,691 homes)	NDCG@3, bill saving per GBP	Top pick right	NDCG@3, carbon per GBP	Top pick right (carbon)
Fixed order, all feasible (brief's baseline)	0.813	37.6%	0.846	45.5%
Engine (payback suitability)	**0.978**	**91.1%**	**0.968**	**89.2%**

- **Ranking** is scored against hidden true saving per pound, not the engine's own numbers: the top pick is right 91% of the time vs 38% for the brief's fixed-order list.
- **Assistant:** on 50 held-out adversarial scenarios, groundedness **100%** and unsupported claims **0%** (all 42 traps blocked); what-if answers on 20 real homes: 60 of 60 grounded after the engine rerun.

Engineering. 186 automated tests (monotone score mappings, feasibility, per-fuel carbon, physics directions, feature parity between training and deployment, grounding). *make build* rebuilds the real data layer, *make train* and *make eval-test* train and score every model, and trained models and processed data install from hosted links in one command each (verified from a clean clone). Every constant in the configuration names its source.